

Tushar Kanti Dey

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Full-Stack Software Engineer — Scalable Systems
Builder

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🐙 [Tusharxhub](https://github.com/Tusharxhub)

Professional Summary

Computer Science undergraduate with strong foundations in data structures, algorithms, and backend system design. Experienced in architecting full-stack applications with structured data models, secure authentication flows, and modular codebases. Focused on clean abstractions, performance optimization, and production-grade engineering practices.

Core Competencies

Frontend Engineering	Next.js, React.js, Tailwind CSS, TypeScript, Component Architecture	Backend Engineering	Node.js, NestJS, Express.js, RESTful API Design, Authentication Systems
Data Systems	PostgreSQL, Supabase, MongoDB, Redis, Schema Design	Computer Science Foundations	Data Structures, Algorithms, OOP, DBMS, Operating Systems
Mobile Systems	React Native, Expo, Rendering Optimization	Engineering Tooling	Git, GitHub, Docker, CI/CD, Vercel

Signature Projects

WebScope – Data Extraction and Processing System

Overview: Designed a backend-driven web data extraction platform focused on structured parsing, normalization, and API delivery.

Engineering Contributions:

- Implemented modular scraping pipelines with structured data transformation layers.
- Designed REST endpoints to securely serve processed data.
- Added validation, error isolation, and request handling safeguards.
- Structured codebase using separation of concerns for maintainability.

Tech Stack: Next.js, Node.js, REST APIs.

Fenix – Real-Time Communication Platform

Overview: Architected a secure real-time video communication system using WebRTC infrastructure.

Engineering Contributions:

- Integrated LiveKit for peer-to-peer video and audio streaming.
- Implemented authenticated session management using Clerk.
- Designed dynamic meeting lifecycle flows with unique room identifiers.
- Structured UI components for predictable rendering and scalability.

Tech Stack: Next.js, LiveKit, Clerk, Tailwind CSS.

Signifiya – Performance-Optimized Mobile Application

Overview: Built a production-ready mobile system emphasizing runtime efficiency and structured API integration.

Engineering Contributions:

- Reduced unnecessary component re-renders through state refactoring.
- Optimized build size via media compression and configuration tuning.
- Implemented secure environment-based API configuration handling.
- Managed Android build lifecycle and deployment workflows.

Tech Stack: React Native, Expo, REST APIs.

Taskhub – Modular Task Management System

Overview: Developed a modular task tracking system with structured CRUD workflows.

Engineering Contributions:

- Designed predictable state transitions for task lifecycle management.
- Built reusable component layers to improve maintainability.
- Structured UI layout for responsive, device-agnostic behavior.
- Deployed optimized production build via Vercel.

Tech Stack: Next.js, React, Tailwind CSS.

Education

2023–Present **B.Tech in Computer Science Engineering**, *Adamas University*, Kolkata, India

Key Strengths

- Strong grasp of algorithmic complexity and structured problem solving.
- Experience designing modular and maintainable system architectures.
- Clear understanding of relational schema design and API contracts.
- Comfortable debugging and optimizing production-level applications.